

Subject Description Form

Subject Code	ITC528
Subject Title	Clothing & Textile Materials & Processes
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	The subject aims to provide students with an in-depth knowledge on the textile and clothing materials and processes. It will further provide an up-to-date exploitation on the functions and applications of advanced textile and clothing materials as well as state-of-the-art processing technologies.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. Understand the principles and interrelationships between fibre, yarn and fabric structures in textiles and apparel materials and processing. 2. Comprehend and describe various quality evaluation and process control systems in modern textiles and apparel manufacturing. 3. Communicate with the professionals in terms of such technical and economical knowledge as applicable to real industrial situation.
Subject Synopsis/ Indicative Syllabus	Textiles 1. Structure, properties and applications of conventional and advanced textile fibers, yarns and fabrics for apparel. 2. Conventional and modern textile processing technologies including advances of textile machines and computer applications (CAD/CAM/CIM/MIS). 3. Quality evaluation and/or Process control of textiles (fiber, yarn and fabric). Clothing 4. New clothing industry. 5. Development in Manufacturing Processes. 6. Flow control.
Teaching/Learning Methodology	The subject consists of lectures, seminars and tutorials. Students are required to do investigations on new materials and processes as well as potential developments, then give presentations and they are encouraged to participate in seminar and tutorial discussions.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			1	2	3
	1. Coursework	50%	✓	✓	✓
	2. Examination	50%	✓	✓	✓
	Total	100%			
	In the continuous assessment, we focus on two major attribute: life long learning and communication skills in addition to the technical knowledge. Lifelong learning can be obtained from group project through searching, selecting, organizing topics and report details, communication skills can be obtained through organising information and written it in report and presenting in the class. At the same time, interaction among group members is a very good way for enhancing their communication skills. In terms of technical skills, students are given some questions to answer as assignments. In the examination, we assess students’ learning performance in terms of defining concepts, explain principles and describe and analyze application examples as well as providing rationals for different textile and clothing materials and processes.				
Student Study Effort Expected	Class contact:				
	▪ Lectures		27 Hrs.		
	▪ Seminars/Tutorials		12 Hrs.		
	Other student study effort:				
	▪ Preparation for presentation		36 Hrs.		
	▪ Assignments		30 Hrs.		
	▪ Total student study effort		105 Hrs.		
	<u>Indicative Reading List and References:</u>				
	<u>Selected Readings</u>				
1. Xiaoming Tao, Smart Fibres, Fabrics and Clothing, Cambridge, England: Woodhead Pub., 2001.					
2. Boca Raton, Horrocks, A. Richard and Anand, Subhash, Handbook of Technical Textiles, Cambridge, England: Woodhead Pub., CRC Press, 2000.					

	3. High Performance Textiles, Elsevier Sequica Sa. 4. Technical Textiles International, Elsevier Advanced Technology. 5. Morton, W.E., Hearle, J.W.S., Physical Properties of Textile Fibres 3rd ed., Butterworths, 1992. 6. Textile Institute, Textile Terms and Definition, 9th ed., 1991. 7. Miller E., Textile Properties and Behaviour in Clothing use, Batsford Technical, 1984. 8. Chuter, A.J., Introduction to Clothing Production Management, BSP, 2nd edition, 1995. 9. Glock, R.E. & Kunz, G.I., Apparel Manufacturing Sewn Product Analysis, 4th Edition, BSP, 2005. 10. Stylios, G., Textile Objective Measurement and Automation in Garment Manufacture (Ed.), E. Horwood, 1991. 11. Carr, H., Carr and Latham's Technology of Clothing Manufacture, 4th ed., Oxford: Blackwell Publishing, 2008. 12. Tyler, D.J., Textile digital printing technologies, Woodhead Publishing, Cambridge, England, 2005 <u>Supplementary Readings</u> 1. High Performance Textiles 2. International Textile Bulletin – fabrics <u>Websites</u> www.merchandisers.hk www.just-style.com http://www.uspto.gov/
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